CSHO16 – Randomized Algorithms

Programming Assignment

Maximum Marks: 20

The task is to design and implement/simulate a Las Vegas algorithm to solve the following problem:

(i) A primary database has ten m-bit strings $U = \{u_1, u_2, \dots, u_{10}\}$, where each $u_i \in \{0,1\}^m$

(ii) A secondary database has ten m-bit strings $V = \{v_1, v_2, \dots, v_{10}\}$, where each $v_i \in \{0,1\}^m$

(iii) The task is to verify whether there exists an integer k ($1 \leq k \leq 10$) such that $u_k = v_k$

The communication complexity of your algorithm (i.e., the number of bits exchanged between the primary and the secondary), cannot exceed $m + b \times log_2 m + c$ (for some positive constants b and c) and being Las Vegas, your algorithm cannot error.

You can choose a distributed environment to implement this algorithm or you may simulate it using concurrent processes or threads on a standalone machine.

Marking scheme and Logistics will be as follows:

(i) 10 Marks for a good algorithm design and proof that the algorithm is Las Vegas and meets the constraint on communication complexity.

(ii) 10 Marks for a good implementation and demo / presentation.

(iii) Algorithm, Proof of Correctness, Code Modules, and Screenshots of sample runs are to be submitted in hard-copy.

(iii) Deadline to Submit: 30 April 2026.

Note: The submission should be your own attempt and originality will be rewarded through marks.

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